TitanTPP: Local and Recurrent History Correction for Event Quantity Prediction

Anonymous Authors

Affiliations withheld for anonymous review

Abstract. Event forecasting requires predicting how much will occur as well as when it will occur. We examine an explicit history path within a causal attention model. TitanTPP's representative CNN+GRU54 architecture corrects attention queries, keys, and values with three observed positions, then adds a recurrent residual between attention blocks. In completed three-seed validation comparisons, CNN+GRU54 attains the lowest mean quantity MAE and RMSE among six common-head encoder adaptations on Taxi and Intermittent, while RAF favors S2P2 by RMSE. A separate 27-condition study compares full correction with bypass and an equal-active-parameter current-only control. Full correction lowers overall quantity RMSE against both controls in every seed on Taxi and Intermittent, but fails this consistency criterion on RAF. Nine Instacart fits yield smaller mean quantity errors for CNN+GRU54 than for MLP8 and MLP12, with a seed-level exception and worse mean overall timing NLL. Large-quantity and timing losses qualify the correction gains. Prototype-retrieval controls remain pending, and these development-validation results do not establish independent generalization.

Keywords: Event forecasting · Quantity prediction · History correction · Temporal point processes

1 Introduction

Intermittent demand alternates between periods without demand and nonzero observations whose sizes can vary. Forecasting therefore requires an account of both the interval to the next demand and its quantity. Croston's method separates demand size from the interval between nonzero observations [5]; later renewal models place this event-based description within a probabilistic framework [24].

Temporal point processes provide a natural representation of event arrivals. Several neural formulations, including RMTPP and Transformer Hawkes Process, predict categorical event types alongside their occurrence times [7,28]. Demand quantity has a numerical magnitude, however, and a type-classification objective does not directly account for the size of a quantity error. The distinction concerns categorical types versus numerical values, rather than discrete versus continuous data: demand counts can themselves be integer-valued. Point processes are not restricted to categorical marks. Deep renewal models already address demand size and arrival intervals, while mixed-attribute event models support numerical attributes [24,6].

Our question concerns the representation of observed history within this numerical prediction task. Recurrence, attention, and continuous-time states already summarize

earlier events [15,26,3]. Yet this does not determine whether a separate path through recent observed states improves quantity prediction. We ask whether an explicit correction adds useful structure beyond a history-aware encoder and a control with the same active parameter budget. All arms receive numerical event inputs and share quantity and timing output heads, so their comparison concerns the correction and retrieval paths.

We investigate this question with TitanTPP. The representative CNN+GRU54 model combines two forms of correction. A local path adjusts attention queries, keys, and values using the current and two preceding observed positions. A recurrent path adds a projected GRU state between the two attention blocks. The suffix 54 denotes the GRU state width. Both paths are residual, allowing the initial model to retain the shared encoder computation. They provide a particular way to reuse history, rather than supplying information unavailable to the causal encoder.

An earlier MLP8 version gives a reason to examine the correction more closely. Across Taxi, Intermittent, and RAF Spare Parts, switching on its explicit previous-state path improved overall quantity RMSE in all 18 comparisons formed by three seeds and two prototype settings. Yet the strongest quantity gains came with worse timing likelihood on Taxi. Moreover, adding prototypes to the corrected model did not improve quantity RMSE consistently across datasets. This pattern motivates two separate questions: the value of an explicit history path, and the value of the rule used to retrieve from a learned prototype bank.

The distinction matters for attribution. Bypassing the correction removes its active trainable capacity, so that comparison alone cannot identify the benefit of explicit history input. We therefore include a control with the same number of active correction parameters that operates only on the current position. For retrieval, we compare uniform averaging, similarity weighting, separately normalized time and quantity queries, and a general learned query. Only the last two introduce query parameters, and their budgets are matched. Within each correction or retrieval study, the arms share the encoder architecture, output-head definitions, and model-selection rule, but are trained independently. Quantity is the primary selection objective, while timing likelihood, large-quantity errors, and computational cost describe the accompanying tradeoffs.

This paper makes three contributions:

1. We describe a local and recurrent correction architecture for next-event quantity prediction within a causal attention model, including the observation rules that prevent access to the target event.
2. We evaluate explicit history correction against bypass and a current-only control with the same active parameter count, while retaining one quantity-based selection rule and reporting timing and large-quantity tradeoffs.
3. We report completed CNN+GRU54 comparisons with six common-head encoder adaptations across three datasets and seeds, followed by a three-seed Instacart architecture comparison. Quantity rankings and timing costs depend on the dataset. Earlier MLP4 and MLP8 findings remain supporting evidence in the appendices.

The external-encoder panel describes performance under a common evaluation contract, and the completed correction controls test the combined local and recurrent path separately. Instacart provides a supplementary architecture comparison. Matched prototype-retrieval controls remain a planned development study.

2 Related Work

Intermittent demand and numerical quantities. Croston models nonzero demand sizes and interdemand intervals separately [5]. The Syntetos–Boylan adjustment addresses bias in its demand-rate estimate [22], whereas Teunter, Syntetos, and Babai model occurrence probability to respond to the risk of inventory obsolescence [23]. Deep renewal processes extend the size–interval representation with learned conditional distributions [24]. DeepAR instead learns autoregressive probabilistic forecasts across related time series [17]. These approaches establish numerical demand prediction as an existing research problem. Our target is the next event's quantity and recorded gap; forecasting demand accumulated over a fixed calendar horizon would require an additional aggregation procedure.

Neural event-history encoders. Hawkes processes describe self-excitation and mutual excitation in event arrivals [9]. Neural TPPs replace fixed history summaries with learned representations and conditional distributions [19]. RMTPP uses a recurrent embedding [7], and the Neural Hawkes Process introduces continuous-time LSTM states [15]. THP, SAHP, and AttNHP apply attention to event histories and irregular timing [28,27,26]; S2P2 develops a continuous-time state-space architecture [3]. These models already represent dependencies across events. We hold a causal attention backbone fixed and test an additional path through observed states. Neither recurrence nor attention alone constitutes the proposed distinction.

Distributions and evaluation targets. Fully neural intensity models parameterize a cumulative hazard [16], while intensity-free models directly specify conditional inter-event distributions [18]. FlexTPP further treats events with mixed discrete and continuous attributes [6]. These formulations preclude a general claim that TPP marks must be categorical. Our quantity head is a numerical regressor, and our time head assigns probability mass to recorded interval bins. Common-head encoder adaptations therefore answer a different comparison from native marked-event likelihoods. Point forecast errors and probabilistic scores also assess different properties [12,8]; we report raw-quantity RMSE and MAE alongside timing NLL at the same quantity-selected checkpoint.

Residual correction. Residual networks establish additive transformation paths [10], and bottleneck adapters apply compact residual modules in transfer learning [11]. ReZero initializes a scalar residual gate at zero [1]. Our zero-initialized local kernels and recurrent output projection preserve the initial base computation through different parameterizations, and the entire event model is trained jointly. The architectural question is whether direct access to recent observed states adds value within a causal attention encoder. The current-only control retains the active parameter count but changes the inputs and operations of the correction path, so the comparison does not isolate parameter count from every possible optimization effect.

Memory and prototype retrieval. Memory networks learn to read from memory representations [20]. Persistent memory adds input-independent key–value entries to attention [21], and product-key memory retrieves learned values from a parameter bank [13].

Titans includes a contextual neural memory whose parameters are updated at test time [2]. In TitanTPP, learned parameters remain fixed during inference; only sequence states and queries depend on the observed prefix. Its persistent attention entries and post-encoder prototypes are separate banks. The retrieval study tests separate normalization of time and quantity queries against a joint learned query with the same summaries and parameter budget. It does not claim the invention of learned memory or top-k retrieval.

3 Method

3.1 Prediction Task and Causal Encoder

Let $\mathcal{H}_j = \{(\Delta_i, q_i)\}_{i \le j}$ be an observed event prefix, where Δ_i is an inter-event interval and $q_i \ge 0$ is an event quantity. The model predicts the recorded interval D_{j+1} and quantity q_{j+1} of the next event. The target position is withheld from the input, and the prediction is read from the last observed position j. Padding is not an event.

Each observed event is embedded as

$$e_i = W_e[\log(1+\Delta_i), \log(1+q_i)]^\top + b_e + p_i, \tag{1}$$

where p_i is a learned position embedding. These two input values are not standardized by training-set moments. The backbone uses Transformer attention [25] in two pre-normalized causal blocks, with hidden width 64, four attention heads, a feed-forward width of 128, and dropout 0.1. Each block also contains 16 learned persistent key–value entries. These entries are distinct from the prototype bank used after the encoder.

Figure 1 shows where the corrections enter. In the first block, the local correction is applied after layer normalization and the query, key, and value projections, before causal attention. The block then completes its attention residual and feed-forward residual. Its output b_i is passed through the recurrent correction before entering the second block. Let h_i denote the resulting second-block output.

3.2 Local and Recurrent History Correction

For a projected query, key, or value sequence X, the local path computes

$$\widetilde{X}_i = X_i + \sum_{\ell=0}^{2} a_{i\ell}\, \kappa_\ell^X \odot X_{s_\ell(i)}. \tag{2}$$

Here $s_0(i) = i$, and $s_1(i), s_2(i)$ are the preceding observed positions in the same correction segment. The mask $a_{i\ell}$ indicates that the corresponding position exists. Each $\kappa_\ell^X \in \mathbb{R}^{64}$ acts channelwise. Applying this three-position operation to queries, keys, and values uses $3 \times 3 \times 64 = 576$ parameters. The operation permits nearby observed states to alter the attention inputs without replacing the attention layer itself.

The recurrent path projects the first-block output into a 54-dimensional state:

$$u_i = Ab_i, \qquad r_i = \mathrm{GRU}(u_i, r_{\mathrm{prev}}), \qquad \widetilde{b}_i = b_i + \mathbf{1}[n_i \ge 2]Br_i. \tag{3}$$

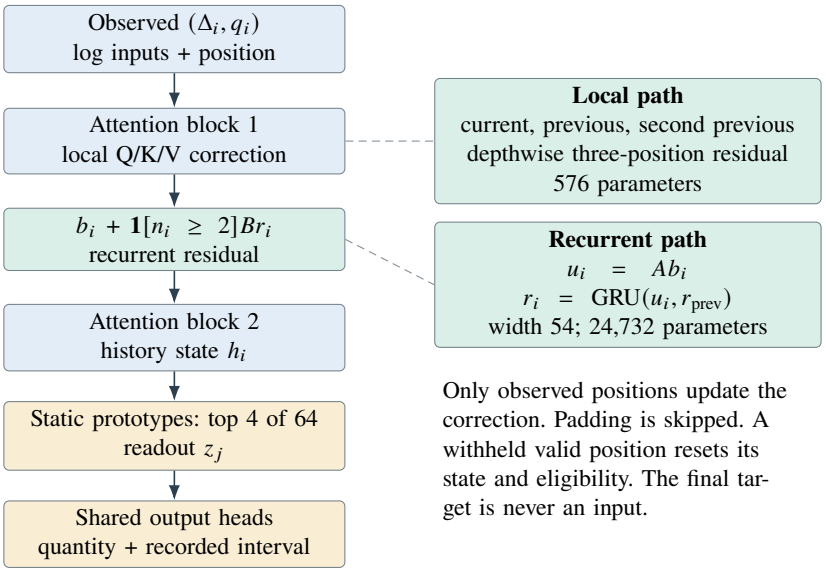


Fig. 1. CNN+GRU54 in TitanTPP. The local correction operates within the first attention block; the recurrent correction operates between blocks. Prototype retrieval and the two output heads follow the encoder. The correction paths contain 25,308 parameters in total.

The GRU follows the gated recurrent formulation [4]; A and B are bias-free projections, and n_i counts observed events in the current correction segment. The residual uses the state *after* processing event i. The recurrent path has 24,732 parameters, including its projections. Its state width is independent of the number of observed events.

Padding neither updates the GRU nor advances the observation count. A valid but withheld position resets the correction state and its eligibility masks. This reset concerns the correction path; it does not change the causal attention mask into a segment-reset mask. The local kernels and recurrent output projection B are initialized to zero. Thus the correction initially leaves the shared encoder computation unchanged, while its input and recurrent parameters can still be learned.

Algorithm 1 collects these operations for one input sequence. Local corrections read the original projected values, while the recurrent residual uses the state after the current observation. Zero initialization of the local kernels and B occurs once before training, rather than at each forward pass.

Algorithm 1 Causal local and recurrent history correction

Require: Observed-prefix embeddings E; valid and observed masks v, o;
causal attention mask $\mathcal{A}$. Withheld target values are excluded.
Ensure: Encoder state h_j at the last observed position $j = \max\{i : o_i = 1\}$.
Construct $s_\ell(i)$ and $a_{i\ell}$ from observed positions.
Skip padding; reset predecessors at valid but unobserved positions.
$(Q, K, V) \leftarrow \text{ProjectQKV}(\text{LayerNorm}(E))$
for $X \in \{Q, K, V\}$ **do**
  $\widetilde{X} \leftarrow X$
  for each observed position i **do**
    $\widetilde{X}_i \leftarrow X_i + \sum_{\ell=0}^{2} a_{i\ell}\, \kappa_\ell^X \odot X_{s_\ell(i)}$
  end for
end for
$b \leftarrow \text{CompleteBlock1}(E, \widetilde{Q}, \widetilde{K}, \widetilde{V}; \mathcal{A})$
Retain persistent entries, attention and feed-forward residuals.
$\widetilde{b} \leftarrow b$; $r \leftarrow 0$; $n \leftarrow 0$
for each position i in sequence order **do**
  if $v_i = 0$ **then**
    continue
  end if
  if $o_i = 0$ **then**
    $r \leftarrow 0$; $n \leftarrow 0$
    continue
  end if
  $r \leftarrow \text{GRU}(Ab_i, r)$
  $n \leftarrow n + 1$
  if $n \geq 2$ **then**
    $\widetilde{b}_i \leftarrow b_i + Br$
  end if
end for
$h \leftarrow \text{Block2}(\widetilde{b}; \mathcal{A})$
return h_j

The correction states are maintained separately for each input sequence. Resetting them leaves the causal attention mask and persistent entries unchanged. The returned state is passed to prototype retrieval and the forecasting heads.

What the controls isolate. The correction-removal variant bypasses both optional paths. Its correction modules remain registered but frozen, contributing zero active trainable correction parameters. The current-only control retains their parameter shapes, initialization, and observation-count masks, but removes direct reads of earlier positions. In Eq. (2), the three bases become X_i, $\tanh(X_i)$, and $\text{GELU}(X_i)$, with the original one-, two-, and three-observation eligibility. For the recurrent component, the current-only control starts from zero at each position and applies two GRU updates using the same current input u_i. Its projected output is used only when $n_i \geq 2$.

The full and current-only paths both contain 25,308 active correction parameters. Their computations nevertheless differ: the current-only control uses nonlinear bases

and repeated current-input updates. It also receives b_i, which already summarizes a prefix through attention. Consequently, this comparison isolates a particular explicit history path against a current-position path; it does not compare a model with history against one without history, or identify the effect of a single lag in isolation.

3.3 Static Prototype Retrieval

After the encoder, a learned bank $M = \{m_k\}_{k=1}^{64}$ supplies a residual to the last observed state h_j. Each slot $m_k \in \mathbb{R}^{64}$ serves as both key and value: its normalized vector is used for scoring, and its raw vector is added to the readout. The slots are fitted with the rest of the model and are not written with observed events at inference time.

Let $\mathrm{norm}(x) = x/\max(\|x\|_2, \epsilon)$ and let S_j contain the four highest-scoring slots. The mean baseline uses cosine scores $s_{jk} = \mathrm{norm}(h_j)^\top \mathrm{norm}(m_k)$ and uniform weights. The weighted variant retains these scores but applies a softmax within the selected four:

$$z_j = h_j + \sum_{k \in S_j} \alpha_{jk} m_k, \qquad \alpha_{jk} = \frac{\exp(s_{jk})}{\sum_{v \in S_j} \exp(s_{jv})}. \tag{4}$$

Mean retrieval instead sets $\alpha_{jk} = 1/4$. Neither variant introduces additional query parameters.

The two learned-query variants receive summaries of the most recent 32 observed positions in the current correction segment, including the current event. Time summaries use standardized $\log(\Delta/s)$; quantity summaries use standardized $\log(1 + q)$, with moments computed from training observations. Each four-dimensional view contains the latest value, mean, population standard deviation, and difference between the latest two eligible values. Nonpositive gaps are excluded from time summaries, whereas zero quantities are retained. The standard deviation and difference require two eligible observations; unavailable summaries are zero. Denote the resulting views by ϕ_t and ϕ_q.

The fixed split-query rule forms

$$v_t = h_j + \phi_t W_t, \qquad v_q = h_j + \phi_q W_q, \tag{5}$$

$$s_{jk}^{\mathrm{fixed}} = \tfrac{1}{2}\,\mathrm{norm}(v_t)^\top \mathrm{norm}(m_k) + \tfrac{1}{2}\,\mathrm{norm}(v_q)^\top \mathrm{norm}(m_k). \tag{6}$$

The two 4×64 projections contain 512 parameters. A budget-matched general-query control uses the same eight summaries, with

$$v_g = h_j + \tfrac{1}{2}[\phi_t, \phi_q] W_g, \qquad s_{jk}^{\mathrm{general}} = \mathrm{norm}(v_g)^\top \mathrm{norm}(m_k), \tag{7}$$

where $W_g \in \mathbb{R}^{8 \times 64}$. Both variants use Eq. (4), temperature one, and zero-initialized query projections. Figure 2 highlights the difference: separate normalization before score averaging versus normalization after the two views have been combined.

The fixed score also has a concatenated dot-product expression:

$$\tfrac{1}{2}[\mathrm{norm}(v_t), \mathrm{norm}(v_q)]^\top [\mathrm{norm}(m_k), \mathrm{norm}(m_k)]. \tag{8}$$

The comparison tests separate query normalization and fixed score mixing. The comparisons share an architecture, not fitted encoder or prototype values: every arm is trained end to end.

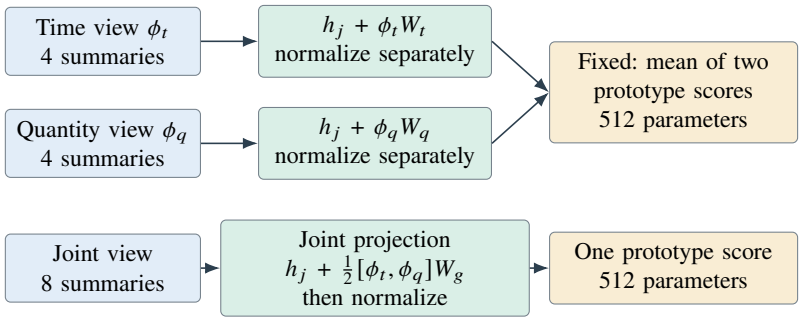


Fig. 2. The planned matched retrieval comparison. Both rules select four of 64 prototypes and use softmax weights. The fixed rule constrains two normalized views to equal score contributions; the general query learns a joint projection before normalization.

3.4 Shared Output Heads and Training Objective

The quantity head predicts in log space and is evaluated in raw units:

$$\widehat{\ell}_j = \mathrm{softplus}(w_q^\top z_j + b_q), \qquad \widehat{q}_{j+1} = \exp(\widehat{\ell}_j) - 1. \tag{9}$$

No inverse training-set standardization is applied. The time head gives

$$\mu_j = w_\mu^\top z_j + b_\mu, \qquad \sigma_j = 10^{-3} + \mathrm{softplus}(w_\sigma^\top z_j + b_\sigma). \tag{10}$$

For a dataset scale $s > 0$, let $T = s\exp(\mu_j + \sigma_j\varepsilon)$ with $\varepsilon \sim \mathcal{N}(0,1)$ and $F_j(x) = \Phi((\log(x/s) - \mu_j)/\sigma_j)$. The recorded interval is $D = \max(1, \mathrm{round}\, T)$, so its probability mass is

$$p_j(d) = \begin{cases} F_j(1.5), & d = 1, \\ F_j(d + 0.5) - F_j(d - 0.5), & d \geq 2. \end{cases} \tag{11}$$

For Instacart, the recorded interval is capped at 30 days: the interior-bin rule applies for $2 \leq d < 30$, and $p_j(30) = 1 - F_j(29.5)$. Training minimizes the sum of recorded-time negative log-likelihood and squared log-quantity error,

$$\mathcal{L} = -\log p_j(D_{j+1}) + \left(\widehat{\ell}_j - \log(1 + q_{j+1})\right)^2. \tag{12}$$

The quantity weight is one, with no extra large-quantity loss term. All compared arms use these heads and this objective.

4 Experimental Design

4.1 Datasets and Model Selection

Taxi, Intermittent, and RAF Spare Parts form the core study. Instacart provides a supplementary architecture comparison. Table 1 gives the eligible target populations and input windows. Lookback is measured in each dataset's native time bins. The inclusive

Table 1. Target populations and sequence construction. q_{large} is a strict lower bound for the large-quantity subset. Scale s is used by the time distribution.

Dataset	Training	Validation	Cap	Lookback	q_{large}	s
Taxi	38,393	8,268	256	168 hours	3449	1
Intermittent	393,824	86,285	256	520 weeks	187	3
RAF	25,779	6,690	84	84 months	200	6
Instacart	1,991,192	503,733	64	52 days	35	7

window ends at the last observed event; the target is appended before truncation to the sequence cap, which includes that target. Large-quantity thresholds are fixed from the training data. Appendix D gives the event definitions and training distributions; Appendix E examines associations across observed-event lags.

The representative experiments use AdamW [14], learning rate 10^{-3}, weight decay 0.01, batch size 128, and gradient clipping at one. Training lasts at least 40 and at most 300 epochs, with patience 40. Each fit retains the first strictly improving finite minimum of overall raw-quantity RMSE on the full validation set. Ties retain the earlier checkpoint. This rule is shared by the earlier common-head panel, the completed MLP8 study, and the new comparisons; neither timing likelihood nor large-quantity error changes the selected epoch.

We evaluate quantity RMSE, quantity MAE, and recorded-time NLL on all validation targets and on the large-quantity subset. All six metrics come from that one checkpoint. Three-seed summaries use the arithmetic mean and sample standard deviation. Time scores are interpreted within a dataset, since the recorded units and distributions differ. Appendix F specifies partitions by quantity and usable history, including how their errors contribute to an aggregate difference.

4.2 Comparisons and Their Scope

Representative model and external encoders. The first comparison places the completed CNN+GRU54 model alongside RMTPP, NHP, THP, SAHP, S2P2, and AttNHP [7,15,28,27,3,26] on the three core datasets. Each external encoder is adapted to the numerical event inputs and common quantity and recorded-time heads. This evaluates forecasting under a common task rather than the models' native marked-event likelihoods. Frozen contracts agree on target identities, input windows, training statistics, loss, optimizer, epoch budget, and checkpoint selection.

CNN+GRU54 contributes nine completed fits, comprising three reused seed-42 runs and six subsequent runs for seeds 52 and 62. The seed-42 Taxi and RAF runs were completed on A100. Intermittent completed epochs 1–12 on A100 and resumed for epochs 13–52 on RTX 5090, retaining the selected epoch-12 checkpoint. Seeds 52 and 62 ran on RTX 5080 for Taxi and RAF and RTX 5090 for Intermittent. The external references came from earlier campaigns, with some software-environment differences. Matching seed labels therefore does not imply common initial parameters or identical training trajectories. This panel describes performance under a shared evaluation contract, while

the following controls address the correction and retrieval mechanisms. Appendix A records the selected epochs and execution provenance.

S2P2 uses two width-64 layers with state size 16, complex dynamics, backward zero-order hold, and relative-time inputs. Its affine-doubling recurrence takes $O(L \log L)$ work for a length-L sequence. AttNHP uses two width-64 layers, one attention head, and 16 time features. Both expose the state immediately after the last observed event to the common heads; neither receives the target event time. Parameter counts and hardware are not matched across all external models, so this comparison does not establish relative structural efficiency. Two nonlearned quantity references repeat the last observed quantity or average quantities within the same input window. Neither supplies a timing distribution.

Completed correction and architecture studies. The correction study compares full CNN+GRU54 correction, correction bypass, and the current-only control on each core dataset and seed. Its 27 distinct conditions are complete. Fifteen conditions completed on A100, three RAF seed-62 conditions on RTX 5080, and nine Intermittent conditions on RTX 5090. The RAF seed-42 full and current-only fits resumed across two physical A100 GPUs after an operational interruption, whereas correction bypass had already completed. Frozen initialization, inputs, loss, optimizer, and checkpoint selection were retained. This migration limits hardware matching for that seed. Appendix G records the selected and final saved epochs.

Within each RTX dataset–seed group, the three arms shared one GPU and Runtime. Seeds nevertheless span hardware and software environments, including cuDNN versions 91900 on RTX 5080 and 92000 on RTX 5090. The nine full-correction fits in this study are kept separate from the nine representative-model fits in Table 2; their means are neither pooled nor substituted between panels.

The supplementary Instacart study also completed nine conditions, comprising MLP8, MLP12, and CNN+GRU54 for seeds 42, 52, and 62. Seeds 42 and 62 ran on RTX 5080, and seed 52 on RTX 5090, with all three architectures sharing one GPU within each seed. The architectures contain 89,859, 96,003, and 102,879 trainable parameters, respectively. Their frozen inputs, common heads, objective, and selection rule agree, but their correction modules and parameter budgets differ. This is an architecture comparison rather than an equal-budget correction ablation. Appendix H retains all six metrics and checkpoint records.

Pending retrieval comparison. The retrieval design compares mean, weighted, fixed, and general-query variants. Fixed and general query each add 512 parameters, whereas the earlier exploratory pilot's general-query control contained 521. A preliminary implementation check on RTX 5080 established 83 bitwise agreements for initialization, synthetic sequences, and the first three updates on a shared Taxi training batch. Its scope does not extend to complete training trajectories.

The completed correction and Instacart studies account for 36 conditions. Retrieval remains pending a common-GPU mean baseline and a new resource contract. The completed A100 full runs are not treated as RTX-matched mean baselines. The additional fit count depends on that baseline contract, so the earlier 63-fit plan does not describe a completed or currently executing campaign.

Decision criteria and computational cost. The correction criterion requires the full model to attain lower overall quantity RMSE than both controls in every seed on the same two or more core datasets. The fixed-retrieval criterion requires corresponding improvements over mean, weighted, and the general-query control. These are descriptive consistency criteria, not significance tests.

Retrieval must also meet a predeclared cost bound relative to mean retrieval on the same GPU. With batch 128, sequence lengths 64 and 256, five warm-up iterations and twenty measured iterations, the allowed increases are 20% for inference time, 30% for training-step time, and 20% for peak allocation. Both lengths must pass. Results from arms above these bounds remain part of the comparison.

5 Results

5.1 CNN+GRU54 against the external encoders

Table 2 compares the representative architecture with the six common-head encoder adaptations. CNN+GRU54 attains the lowest mean quantity MAE and RMSE on Taxi and Intermittent. Relative to RMTPP, the strongest external reference by mean RMSE on both datasets, its RMSE is lower by 16.54% and 32.17%, respectively. These are comparisons of completed runs from separate campaigns under the shared evaluation contract in Section 4.

On RAF, S2P2 attains lower mean RMSE, 33.9947 versus 34.1841 for CNN+GRU54. RMTPP also attains lower mean MAE, 9.1362 versus 9.2062. The quantity ranking therefore depends on the dataset. The supplementary Instacart architecture comparison appears separately in Section 5.3, without extending this external-encoder panel to a new campaign.

Timing scores limit the quantity advantage. On Taxi, CNN+GRU54 records time NLL 1.4684, compared with 0.6516 for NHP. On Intermittent, the corresponding scores are 0.3025 for CNN+GRU54 and 0.2349 for S2P2. NHP also records lower time NLL on RAF, 3.3889 versus 3.5464. Each score comes from the same checkpoint as its row's quantity errors, so none substitutes an independently selected timing checkpoint.

The nonlearned references provide a further check. CNN+GRU54 attains lower RMSE than both rules on Taxi and RAF. On Intermittent, repeating the last quantity yields lower MAE, 0.5647 versus 0.7022, and slightly lower RMSE, 1.7078 versus 1.7296. Thus the lower errors against learned encoder adaptations do not establish superiority over every simple predictor. Appendix B retains the numerical reference values, and Appendix A reports all six CNN+GRU54 metrics, including the large-quantity subsets.

5.2 Additional effect of explicit history correction

Table 3 reports the completed 27-condition correction study. Against both correction bypass and current-only, full correction lowers overall quantity RMSE in all three seeds on Taxi and Intermittent. These two datasets meet the predeclared consistency criterion. Although full correction also attains lower mean RMSE on RAF, seed 52 favors both

Table 2. Validation performance of TitanTPP (CNN+GRU54) and six common-head encoder adaptations. Entries are mean ± sample SD over seeds 42, 52, and 62, at each fit's quantity-RMSE-selected checkpoint. Data, training budgets, and selection rules agree, but campaigns and hardware differ. The CNN+GRU54 row includes three reused seed-42 fits and six subsequent fits. Bold marks the lowest mean in each dataset and metric.

Model	Quantity MAE	Quantity RMSE	Time NLL
Taxi			
CNN+GRU54	**25.2797 ± 0.7099**	**79.0628 ± 2.6668**	1.4684 ± 0.6504
RMTPP	28.5124 ± 0.6344	94.7283 ± 3.4006	1.5798 ± 0.9400
THP	32.4095 ± 0.8144	107.5977 ± 4.5040	0.6629 ± 0.0174
NHP	94.2694 ± 1.8752	318.8148 ± 5.3011	**0.6516 ± 0.0021**
SAHP	34.2260 ± 0.4539	120.5273 ± 1.9996	0.6848 ± 0.0146
S2P2	33.0231 ± 0.5569	106.3540 ± 1.9135	0.6658 ± 0.0242
AttNHP	35.9811 ± 1.6088	127.1290 ± 8.9919	0.6876 ± 0.0194
Intermittent			
CNN+GRU54	**0.7022 ± 0.0122**	**1.7296 ± 0.0477**	0.3025 ± 0.0181
RMTPP	0.8949 ± 0.0494	2.5499 ± 0.1604	0.2748 ± 0.0047
THP	0.8993 ± 0.0661	2.7534 ± 0.1784	0.2911 ± 0.0045
NHP	4.5188 ± 0.1706	12.8755 ± 1.0405	0.4613 ± 0.0969
SAHP	1.8619 ± 0.0899	7.0298 ± 0.7673	0.3448 ± 0.0138
S2P2	0.9359 ± 0.0086	2.5758 ± 0.0766	**0.2349 ± 0.0189**
AttNHP	1.0413 ± 0.0938	3.0799 ± 0.2982	0.2946 ± 0.0569
RAF			
CNN+GRU54	9.2062 ± 0.0741	34.1841 ± 0.2334	3.5464 ± 0.0659
RMTPP	**9.1362 ± 0.0557**	34.5503 ± 0.2506	3.5191 ± 0.1463
THP	9.3014 ± 0.1819	34.7445 ± 0.3204	3.3928 ± 0.0187
NHP	9.1793 ± 0.0882	36.2731 ± 0.3186	**3.3889 ± 0.0556**
SAHP	9.2332 ± 0.0209	35.4193 ± 0.2130	3.7415 ± 0.0745
S2P2	9.2730 ± 0.0517	**33.9947 ± 0.1261**	3.4187 ± 0.0284
AttNHP	9.1917 ± 0.0434	34.8766 ± 0.0676	3.9707 ± 0.0609

controls: full RMSE is 34.4012, compared with 34.3415 for bypass and 34.3349 for current-only. RAF therefore fails the seedwise criterion. Across the 18 full-versus-control comparisons, 16 favor full correction by overall RMSE.

The quantity advantage carries costs. Full correction raises overall time NLL in 11 of the 18 comparisons, including all six Taxi comparisons. Large-quantity RMSE and MAE each worsen in ten comparisons. On Intermittent, for example, full correction attains mean overall RMSE 1.6957 versus 1.7409 for bypass, but large-quantity RMSE rises to 8.2968 versus 7.8753. On RAF, the corresponding large-quantity means are 327.6237 and 324.7870. These metrics inherit the same quantity-selected checkpoint. Appendix G retains all six metrics and the adverse seedwise differences rather than selecting a separate favorable timing or tail epoch.

The results support the tested combined correction path within development validation. Both full and current-only contain 25,308 active correction parameters, yet their input operations and recurrent depth differ. Because the current contextual state already

Table 3. Completed CNN+GRU54 correction study. Entries are mean ± sample SD over seeds 42, 52, and 62 at each fit's earliest minimum overall quantity-RMSE checkpoint. These full-model fits form a separate campaign from Table 2. Bold marks the lowest mean within a dataset and column. RAF seed 42 includes physical A100 migration.

Arm	Quantity MAE	Quantity RMSE	Time NLL
Taxi			
Full	**25.6347 ± 1.0220**	**80.4089 ± 2.9830**	1.2721 ± 0.1648
Correction bypass	28.2757 ± 0.4683	87.6117 ± 1.4166	0.9544 ± 0.1116
Current-only	28.1968 ± 0.3279	88.0008 ± 1.3758	**0.8206 ± 0.2261**
Intermittent			
Full	**0.7057 ± 0.0148**	**1.6957 ± 0.0423**	0.4097 ± 0.1818
Correction bypass	0.7449 ± 0.0276	1.7409 ± 0.0279	**0.3430 ± 0.0405**
Current-only	0.7532 ± 0.0227	1.7562 ± 0.0574	0.3479 ± 0.0244
RAF			
Full	**9.2021 ± 0.0801**	**34.1711 ± 0.2121**	3.5481 ± 0.0634
Correction bypass	9.2676 ± 0.0325	34.2964 ± 0.0846	3.5891 ± 0.1286
Current-only	9.3667 ± 0.1663	34.3109 ± 0.1010	**3.4693 ± 0.1006**

includes prefix history, this control does not remove history from the entire model. The comparison also does not identify the contribution of the CNN or GRU separately. RAF seed 42's physical GPU migration and cross-seed Runtime differences limit hardware matching. Concurrent training durations do not establish structural efficiency.

The earlier MLP8 study remains a separate observation in Appendix C. Its 18 correction gains concern that MLP path and are not substituted for the CNN+GRU54 controls.

5.3 Supplementary Instacart architecture comparison

The nine completed Instacart fits extend the architecture comparison to basket quantities on 503,733 validation targets. CNN+GRU54 attains the lowest mean overall quantity RMSE and MAE among the three variants in Table 4. Its mean RMSE is 5.8774, compared with 5.8854 for MLP8 and 5.8801 for MLP12. The gaps are small relative to the reported seed dispersions. For seed 52, MLP12 attains lower overall and large-quantity RMSE and MAE than CNN+GRU54. Each quantity metric consequently favors CNN+GRU54 in five of the six seedwise comparisons, rather than in every comparison.

For the 6,036 targets with quantity above 35, CNN+GRU54 attains mean RMSE 24.4380 and MAE 21.7822, compared with RMSE 24.6355 and 24.6683 for MLP8 and MLP12. Its large-quantity time NLL is lower in all six seedwise comparisons. Mean overall time NLL, however, rises to 2.8108 from 2.8075 and 2.8072. The seed-62 CNN+GRU54 fit incurs higher overall time NLL than both MLP variants. Appendix H reports both populations at the same selected checkpoints.

Instacart therefore supplies a modest architecture-level quantity advantage with explicit exceptions. Different parameter counts prevent attribution to an equal-budget

Table 4. Completed Instacart architecture comparison on 503,733 validation targets. Each architecture contributes three seeds and all columns inherit its quantity-RMSE-selected checkpoint. Entries are mean ± sample SD. Bold marks the lowest mean in each column. These variants have different parameter counts.

Architecture	Quantity MAE	Quantity RMSE	Time NLL
MLP8	3.9910 ± 0.0028	5.8854 ± 0.0135	2.8075 ± 0.0040
MLP12	3.9906 ± 0.0025	5.8801 ± 0.0169	**2.8072 ± 0.0050**
CNN+GRU54	**3.9872 ± 0.0041**	**5.8774 ± 0.0095**	2.8108 ± 0.0090

history intervention, and this study does not test prototype retrieval. Its results are not merged with the earlier MLP4 external comparison in Appendix B.

5.4 Additional effect of prototype retrieval

The matched CNN+GRU54 retrieval comparison remains pending a common-GPU baseline and resource contract. It compares mean, weighted, fixed, and general-query retrieval, retaining the same encoder and output-head definitions. A fixed-query gain over mean alone would not identify the value of separating time and quantity queries, because weighting and general query learning remain alternative explanations. No completed three-dataset result for this matched comparison is reported here.

6 Discussion

The completed CNN+GRU54 results support two bounded findings. In the external-encoder panel, its mean quantity errors are lower than those of the six tested common-head adaptations on Taxi and Intermittent, while RAF favors S2P2 by mean RMSE. The separate correction study favors the full path against both controls in every seed on Taxi and Intermittent, with a seed-level failure on RAF. Retaining one representative architecture across datasets avoids a different best-variant choice for each result.

This correction finding concerns the combined local and recurrent computation. Equal active parameter counts narrow a capacity explanation, but current-only changes the nonlinear bases and repeated-input updates. Its current hidden state also contains prefix history. The controls consequently distinguish the tested computational paths without identifying a unique lag or separating CNN and GRU effects. Earlier MLP8 results provide context rather than the mechanism evidence for CNN+GRU54.

Instacart supplies a smaller quantity difference. CNN+GRU54's lower mean errors coexist with a seed-52 MLP12 advantage and higher mean overall timing NLL. These variants differ in parameter count, and their limited mean gaps do not establish a statistically significant architectural advantage. Training-history length and event-lag associations in the appendices describe data characteristics rather than causes of the observed model ranking.

Quantity gains do not imply uniformly better forecasting. The correction study includes 11 overall timing losses and ten losses each for large-quantity RMSE and MAE

among 18 comparisons. All scores come from quantity-selected checkpoints. Reporting these exceptions prevents the primary quantity criterion from concealing timing and rare-target costs.

The retrieval hypothesis remains distinct. Fixed split-query scoring can be expressed as a concatenated dot product, and a future advantage over the budget-matched general query would support the tested separate-normalization constraint. No completed CNN+GRU54 retrieval effect is established here. A100 full fits also cannot substitute for an RTX common-GPU mean baseline.

The external panel and correction study combine preserved and separately trained runs across hardware. RAF seed 42's correction trajectories include physical A100 migration, and seed summaries span Runtime differences. Task and selection agreement permit descriptive comparisons but do not establish hardware efficiency. All reported performance derives from validation evidence already encountered during research. CPU binary re-inference audits and an independently unexposed evaluation remain outstanding, and three-seed summaries do not establish statistical significance.

7 Conclusion

TitanTPP augments a causal attention encoder with local and recurrent history correction. Completed CNN+GRU54 runs attain lower mean quantity errors than six common-head encoder adaptations on Taxi and Intermittent, while S2P2 attains lower mean RMSE on RAF. The completed correction study supports the full path against bypass and an equal-active-parameter current-only control in every seed on Taxi and Intermittent. RAF fails the same criterion, and timing and large-quantity losses remain substantial qualifications. Instacart supplies a smaller mean quantity advantage over MLP8 and MLP12 with a seed-level exception and worse mean overall timing NLL. Matched retrieval and independent evaluation remain pending. These results support a bounded development-validation claim for the combined correction path.

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A Completed CNN+GRU54 records

Table 5 identifies the nine completed representative-model fits used in Table 2. Seeds 42, 52, and 62 each contribute once per dataset. Seed-42 reuse preserves the original checkpoint selection, including the resumed Intermittent trajectory. These representative-model records remain separate from the full-correction campaign summarized in Appendix G.

Table 5. Selected and final saved epochs for completed CNN+GRU54 fits. A100 runs are preserved originals. Intermittent seed 42 completed epochs 1–12 on A100 and 13–52 on RTX 5090, retaining the selected epoch-12 checkpoint.

Dataset	Seed	Selected	Final	Execution
Taxi	42	126	166	A100, reused
	52	157	197	RTX 5080
	62	54	94	RTX 5080
Intermittent	42	12	52	A100 → RTX 5090
	52	11	51	RTX 5090
	62	13	53	RTX 5090
RAF	42	9	49	A100, reused
	52	5	45	RTX 5080
	62	20	60	RTX 5080

Table 6 retains the overall and large-quantity metrics at the same selected checkpoints. Large subsets contain 79 Taxi, 1,141 Intermittent, and 50 RAF targets. Their membership is fixed by the training-derived thresholds. The three seeds repeat this evaluation population rather than tripling its number of targets.

Table 6. CNN+GRU54 validation metrics, mean ± sample SD over three seeds. Each fit contributes the same selected checkpoint to both populations. Very small time NLLs retain scientific notation.

Dataset	Population	Quantity MAE	Quantity RMSE	Time NLL
Taxi	All	25.2797 ± 0.7099	79.0628 ± 2.6668	1.4684 ± 0.6504
	Large	303.9219 ± 29.0428	392.1246 ± 24.7344	1.367×10^{-19} $\pm 2.368\times10^{-19}$
Intermittent	All	0.7022 ± 0.0122	1.7296 ± 0.0477	0.3025 ± 0.0181
	Large	6.6967 ± 0.2454	8.6809 ± 0.1492	9.239×10^{-5} $\pm 1.527\times10^{-4}$
RAF	All	9.2062 ± 0.0741	34.1841 ± 0.2334	3.5464 ± 0.0659
	Large	281.5753 ± 8.7931	328.3920 ± 7.8560	3.4953 ± 0.2210

B Earlier MLP4 common-head comparison

This appendix preserves the earlier four-dataset comparison with TitanTPP-MLP4. Its correction comprises eight width-four MLP branches and 6,144 nominal parameters. These checkpoints differ from both MLP8 and CNN+GRU54. The historical seven-model panel contains 84 fits, and its six external encoder references on the core datasets are also retained in Table 2.

B.1 Overall performance and nonlearned references

Table 7 reports the completed four-dataset comparison. On Taxi and Intermittent, TitanTPP-MLP4 has lower quantity MAE and RMSE than all six encoder adaptations in every paired seed. These are 36 model–dataset–seed comparisons for each quantity metric. Relative to RMTPP, which has the lowest mean quantity errors among the six alternatives on both datasets, mean RMSE is lower by 15.85% on Taxi and 34.39% on Intermittent; mean MAE is lower by 10.13% and 21.72%, respectively.

The result is narrower on RAF. MLP4 has the lowest mean RMSE, but its advantage over S2P2 is only 0.13%, with improvements in two of three paired seeds. RMTPP has lower mean MAE. On Instacart, S2P2 has the lowest mean RMSE and timing NLL, while RMTPP has the lowest mean MAE. MLP4 has 0.52% higher mean RMSE than S2P2, and both S2P2 and AttNHP have lower RMSE in all three paired seeds. These differences rule out a uniform quantity ranking across the four datasets.

Timing provides a further distinction. NHP has the lowest mean time NLL on Taxi and RAF, and S2P2 does so on Intermittent and Instacart. For example, MLP4's Taxi NLL is 1.0387, compared with 0.6516 for NHP; on Intermittent, it is 0.4971, compared with 0.2349 for S2P2. The quantity advantage therefore coexists with worse timing scores at the quantity-selected checkpoints.

Table 8 puts these quantity results alongside the two nonlearned rules. MLP4 has lower RMSE than both on all four datasets. On Intermittent, however, repeating the last quantity gives lower MAE, 0.5647 versus 0.7005, while its RMSE is 1.7078 versus 1.6730. This disagreement between MAE and RMSE is material: outperforming the learned encoders does not mean that the model improves every measure over a simple rule.

These results describe the archived MLP4 variant under common heads. The representative CNN+GRU54 comparison is reported separately in Table 2.

Table 7. Historical development-validation comparison with numerical-input, common-head adaptations of the external encoders. Values are mean ± sample standard deviation over seeds 42, 52, and 62; all three metrics use the same RMSE-selected checkpoint within each run. TitanTPP (MLP4) denotes the earlier eight-branch, width-four correction (6,144 parameters), not CNN+GRU54. Bold marks the lowest mean in each dataset and metric within this panel.

Model	Quantity MAE	Quantity RMSE	Time NLL
Taxi			
TitanTPP (MLP4)	**25.6237 ± 0.8320**	**79.7111 ± 2.8260**	1.0387 ± 0.2843
RMTPP	28.5124 ± 0.6344	94.7283 ± 3.4006	1.5798 ± 0.9400
THP	32.4095 ± 0.8144	107.5977 ± 4.5040	0.6629 ± 0.0174
NHP	94.2694 ± 1.8752	318.8148 ± 5.3011	**0.6516 ± 0.0021**
SAHP	34.2260 ± 0.4539	120.5273 ± 1.9996	0.6848 ± 0.0146
S2P2	33.0231 ± 0.5569	106.3540 ± 1.9135	0.6658 ± 0.0242
AttNHP	35.9811 ± 1.6088	127.1290 ± 8.9919	0.6876 ± 0.0194
Intermittent			
TitanTPP (MLP4)	**0.7005 ± 0.0541**	**1.6730 ± 0.0819**	0.4971 ± 0.2095
RMTPP	0.8949 ± 0.0494	2.5499 ± 0.1604	0.2748 ± 0.0047
THP	0.8993 ± 0.0661	2.7534 ± 0.1784	0.2911 ± 0.0045
NHP	4.5188 ± 0.1706	12.8755 ± 1.0405	0.4613 ± 0.0969
SAHP	1.8619 ± 0.0899	7.0298 ± 0.7673	0.3448 ± 0.0138
S2P2	0.9359 ± 0.0086	2.5758 ± 0.0766	**0.2349 ± 0.0189**
AttNHP	1.0413 ± 0.0938	3.0799 ± 0.2982	0.2946 ± 0.0569
RAF			
TitanTPP (MLP4)	9.2506 ± 0.0855	**33.9507 ± 0.0588**	3.5446 ± 0.0669
RMTPP	**9.1362 ± 0.0557**	34.5503 ± 0.2506	3.5191 ± 0.1463
THP	9.3014 ± 0.1819	34.7445 ± 0.3204	3.3928 ± 0.0187
NHP	9.1793 ± 0.0882	36.2731 ± 0.3186	**3.3889 ± 0.0556**
SAHP	9.2332 ± 0.0209	35.4193 ± 0.2130	3.7415 ± 0.0745
S2P2	9.2730 ± 0.0517	33.9947 ± 0.1261	3.4187 ± 0.0284
AttNHP	9.1917 ± 0.0434	34.8766 ± 0.0676	3.9707 ± 0.0609
Instacart			
TitanTPP (MLP4)	3.9916 ± 0.0026	5.8827 ± 0.0048	2.8071 ± 0.0041
RMTPP	**3.9852 ± 0.0052**	5.8775 ± 0.0213	2.8051 ± 0.0048
THP	4.0013 ± 0.0025	5.8582 ± 0.0253	2.8056 ± 0.0016
NHP	4.4923 ± 0.0304	6.7613 ± 0.0538	2.8132 ± 0.0080
SAHP	3.9979 ± 0.0099	5.8776 ± 0.0067	2.8018 ± 0.0013
S2P2	3.9987 ± 0.0063	**5.8523 ± 0.0150**	**2.7970 ± 0.0014**
AttNHP	3.9958 ± 0.0021	5.8648 ± 0.0194	2.8080 ± 0.0028

Table 8. Historical validation comparison with nonlearned quantity rules on the same targets and observed windows. TitanTPP (MLP4) entries are three-seed means; deterministic rules have one result, no seed standard deviation, and no duration likelihood.

Dataset	Predictor	MAE	RMSE
Taxi	TitanTPP (MLP4)	25.6237	79.7111
	Last quantity	41.1951	134.3931
	Window mean	120.3771	368.7843
Intermittent	TitanTPP (MLP4)	0.7005	1.6730
	Last quantity	0.5647	1.7078
	Window mean	4.5537	13.7821
RAF	TitanTPP (MLP4)	9.2506	33.9507
	Last quantity	11.3136	44.5742
	Window mean	10.8564	36.8499
Instacart	TitanTPP (MLP4)	3.9916	5.8827
	Last quantity	4.9674	7.3473
	Window mean	4.0507	5.9030

Table 9. Completed MLP8 study: overall validation quantity RMSE (mean ± sample SD over three seeds; lower is better). Bold marks the lowest mean within each dataset. These are not CNN+GRU54 results.

Arm	Taxi	Intermittent	RAF
P_0H_0	88.977 ± 0.501	1.777 ± 0.065	34.242 ± 0.147
P_1H_0	90.509 ± 0.521	1.721 ± 0.045	34.230 ± 0.127
P_0H_1	**77.377** ± 1.822	**1.617** ± 0.009	33.979 ± 0.045
P_1H_1	77.735 ± 0.681	1.664 ± 0.009	**33.938** ± 0.071
P_FH_1	79.255 ± 1.230	1.759 ± 0.124	34.125 ± 0.074
P_1H_C	88.846 ± 1.267	1.728 ± 0.061	34.273 ± 0.327

C Earlier MLP8 correction study

C.1 Architecture and controls

The initial study varies prototype use (P_0/P_1) and MLP correction (H_0/H_1) in a 2×2 comparison. It places an MLP correction between the first and second attention blocks, with no local Q/K/V correction; prototype retrieval follows the second block. Thus it differs from both correction paths in CNN+GRU54. Eight bias-free branches each map the concatenated current and immediately preceding observed states through $128 \to 8 \to 64$, and their outputs are averaged. The branches become eligible after more than 1, 2, 4, 8, 16, 32, 64, or 128 observations. These are eligibility thresholds, not eight different history lags.

The correction contains 12,288 parameters; the 64-slot prototype bank contains 4,096. Two additional controls test alternatives to these components. P_1H_C replaces the correction with eight current-state branches of width 12, matching its parameter count. P_FH_1 replaces the prototype residual with a bias-free $64 \to 32 \to 64$ feed-forward residual, also using 4,096 parameters. All six arms retain the backbone's persistent attention entries. The six arms, three datasets, and seeds 42, 52, and 62 produce 54 completed fits.

These budget controls have limits. The feed-forward substitute differs in activation and initialization from prototype retrieval. Equal parameter counts do not imply equal computation. On RAF, the sequence cap prevents the eighth MLP branch from becoming eligible, leaving seven potentially active branches. These details constrain what can be attributed to the nominal budgets.

C.2 Quantity gains and parameter controls

Table 9 reports the complete six-arm quantity comparison. Adding the MLP correction reduces overall RMSE in all 18 paired comparisons: three datasets, three seeds, and two prototype settings. Without prototypes, the reductions in the means are 13.04% on Taxi, 9.01% on Intermittent, and 0.77% on RAF. With prototypes, the corresponding reductions are 14.11%, 3.28%, and 0.85%. The direction is consistent, but the magnitude is strongly dataset-dependent.

The parameter controls provide a more selective result. P_1H_1 beats both P_FH_1 and P_1H_C in all three seeds on Taxi and RAF. On Intermittent, it beats each control in only two of three seeds. Thus the MLP8 study satisfies its two-dataset consistency criterion, while leaving an exception on Intermittent. The large Taxi gap against P_1H_C is compatible with useful explicit previous-state input in this MLP architecture; the control does not establish which part of the CNN+GRU path would account for such an effect.

Prototype addition yields a different pattern. With the correction already present, P_1H_1 has 0.46% higher mean RMSE than P_0H_1 on Taxi and 2.92% higher RMSE on Intermittent. On RAF it improves the mean by 0.12%. The completed evidence therefore favors explicit correction more consistently than prototype addition. Choosing a different best variant for each dataset would conceal this distinction and would not define a single representative model.

C.3 Timing, large quantities, and cost

Quantity gains are not uniform gains across outputs. On Taxi, adding correction raises overall time NLL from 0.818 to 1.253 without prototypes and from 0.725 to 1.229 with prototypes. These are increases of 53.17% and 69.64% in the means; timing NLL worsens in every seed under both settings. The checkpoints were selected for quantity RMSE, so this is a tradeoff at the quantity-selected model rather than a comparison of separately optimized timing checkpoints.

Large-quantity behavior also qualifies the overall result. The Taxi full model has large-quantity RMSE 376.660, compared with 366.709 for its feed-forward prototype substitute, despite lower overall RMSE. On RAF, adding correction slightly reduces overall RMSE but increases large-quantity MAE by 1.27% without prototypes and 0.94% with prototypes. Its large-quantity time NLL rises by 3.64% and 3.85%, respectively. The large subsets contain only 79 Taxi targets and 50 RAF targets, compared with 1,141 on Intermittent. Table 10 retains all six metrics, including these unfavorable outcomes.

The historical technical checks do not show that prototype retrieval is cheaper than its parameter-matched substitute. On RTX 5080, the measured step time for P_1H_1 is 36.46 ms versus 25.34 ms for P_FH_1 at length 256, and 14.15 ms versus 11.49 ms at length 84. On RTX 5090 at length 256, the corresponding times are 23.11 and 17.74 ms. These checks use the median of three synthetic optimizer steps, without discarded warm-up, and include gradient-copy and verification work. They are limited technical measurements, not pure throughput or inference benchmarks, and do not implement the new retrieval cost protocol.

The completed Instacart architecture comparison is reported in Section 5.3 and Appendix H. It supersedes the earlier single-fit monitoring observation without replacing the core-dataset MLP8 study preserved here.

C.4 Full validation metrics

All entries in Table 10 are means and sample standard deviations over seeds 42, 52, and 62. Each fit contributes metrics from its first minimum-overall-quantity-RMSE checkpoint. The large subsets use the strict training-derived thresholds in Table 1. Values are rounded to three decimals except for time NLLs below 0.001, which use

scientific notation with the standard deviation on the line beneath the mean. No metric in this table comes from the representative CNN+GRU experiment.

Table 10. Completed MLP8 study: all six validation metrics (mean ± sample SD). Lower is better in every column. The large-quantity subset is shown beside the full population rather than used for independent checkpoint selection.

	All targets			Large-quantity targets		
Arm	RMSE	MAE	Time NLL	RMSE	MAE	Time NLL
Taxi: 8,268 targets; 79 large						
P_0H_0	88.977 ± 0.501	28.313 ± 0.096	0.818 ± 0.191	419.317 ± 53.142	336.399 ± 45.621	1.21×10^{-30} $\pm2.09\times10^{-30}$
P_1H_0	90.509 ± 0.521	28.755 ± 0.453	0.725 ± 0.067	431.690 ± 19.168	346.009 ± 21.136	6.36×10^{-17} $\pm1.10\times10^{-16}$
P_0H_1	77.377 ± 1.822	25.180 ± 0.491	1.253 ± 0.125	385.604 ± 27.258	311.795 ± 29.080	9.77×10^{-78} $\pm1.69\times10^{-77}$
P_1H_1	77.735 ± 0.681	25.040 ± 0.146	1.229 ± 0.160	376.660 ± 10.552	296.101 ± 11.577	5.51×10^{-62} $\pm9.54\times10^{-62}$
P_FH_1	79.255 ± 1.230	25.475 ± 0.205	1.155 ± 0.373	366.709 ± 26.906	287.777 ± 23.170	1.58×10^{-43} $\pm2.74\times10^{-43}$
P_1H_C	88.846 ± 1.267	28.422 ± 0.295	1.001 ± 0.322	398.429 ± 29.267	314.315 ± 23.846	1.82×10^{-25} $\pm3.15\times10^{-25}$
Intermittent: 86,285 targets; 1,141 large						
P_0H_0	1.777 ± 0.065	0.751 ± 0.042	0.366 ± 0.016	8.121 ± 0.462	6.406 ± 0.382	0.035 ± 0.046
P_1H_0	1.721 ± 0.045	0.735 ± 0.048	0.371 ± 0.050	8.000 ± 0.165	6.163 ± 0.145	1.32×10^{-4} $\pm2.28\times10^{-4}$
P_0H_1	1.617 ± 0.009	0.669 ± 0.015	0.384 ± 0.041	7.510 ± 0.441	5.561 ± 0.283	0.004 ± 0.007
P_1H_1	1.664 ± 0.009	0.722 ± 0.009	0.370 ± 0.059	7.711 ± 0.125	5.986 ± 0.290	0.016 ± 0.028
P_FH_1	1.759 ± 0.124	0.745 ± 0.105	0.356 ± 0.068	8.911 ± 0.805	6.481 ± 0.575	0.387 ± 0.645
P_1H_C	1.728 ± 0.061	0.710 ± 0.023	0.551 ± 0.350	8.315 ± 0.473	6.541 ± 0.402	9.908 ± 17.131
RAF: 6,690 targets; 50 large						
P_0H_0	34.242 ± 0.147	9.270 ± 0.057	3.592 ± 0.112	326.556 ± 6.460	279.427 ± 7.680	3.455 ± 0.150
P_1H_0	34.230 ± 0.127	9.261 ± 0.062	3.596 ± 0.112	326.405 ± 6.068	279.348 ± 7.255	3.447 ± 0.145
P_0H_1	33.979 ± 0.045	9.259 ± 0.099	3.554 ± 0.078	327.105 ± 1.589	282.972 ± 2.416	3.581 ± 0.220
P_1H_1	33.938 ± 0.071	9.266 ± 0.092	3.552 ± 0.081	326.227 ± 2.112	281.968 ± 2.766	3.580 ± 0.223
P_FH_1	34.125 ± 0.074	9.189 ± 0.031	3.470 ± 0.029	329.715 ± 0.401	285.503 ± 0.716	3.336 ± 0.068
P_1H_C	34.273 ± 0.327	9.291 ± 0.084	3.548 ± 0.075	327.082 ± 4.843	282.363 ± 6.635	3.445 ± 0.144

D Data definitions and training distributions

Taxi events record pickup counts in active spatial-cell hours. Intermittent events contain positive site–part order quantities, and RAF events contain positive monthly spare-parts demands. Instacart aggregates product rows for each user on the same recorded activity day, so its quantity measures basket size rather than physical units. Empty calendar periods extend the interval between events instead of creating zero-quantity targets. The recorded units are hours, weeks, months, and days, respectively. Instacart intervals are top-coded at 30 days.

Table 11 describes the frozen training populations. Quantity summaries include every training event, whereas gap and usable-history summaries exclude each series' first event because it has no predecessor for prediction. Quantiles select the nearest observed value. Each summary weights events or prediction targets equally within its population, rather than weighting series equally.

Usable history counts the observed predecessor events retained for a target after both window restrictions. The inclusive recorded-time window ends at the last observed event. Appending the target before sequence truncation reserves one position for it, so at most the sequence cap minus one observed events remain. Neither the target nor padding contributes to usable history. Table 1 specifies the windows and caps. The train descriptions therefore characterize the same input contract across the MLP and CNN+GRU studies, although those architectures process the retained events differently.

Table 11. Training-only event characteristics. Quantity summaries include all training events; gap, usable-history, and adjacent-pair summaries describe targets after excluding the first event of each series. Correlation is defined in Appendix E.

Characteristic	Taxi	Intermittent	RAF	Instacart
Time unit	Hour	Week	Month	Day
Training series	131	5,000	5,000	206,209
Training event rows	38,524	398,824	30,779	2,197,401
Training targets	38,393	393,824	25,779	1,991,192
Quantity: median	7	2	2	8
Quantity: 95th percentile	1,562	46	60	25
Quantity: 99th percentile	3,449	187	200	35
Quantity: maximum	6,322	477	2,062	175
Recorded gap: median	1	3	6	7
Recorded gap: 95th percentile	5	5	23	30
Usable history: median	129	40	3	5
Usable history: 95th percentile	168	144	8	14
History ≤ 7 events (%)	2.41	8.89	93.58	72.13
Within-series log correlation	0.749	0.928	-0.105	0.037
Equal adjacent quantities (%)	20.33	61.99	33.30	10.24

Figure 3 indicates substantial differences in available history. The median spans 129 observed events on Taxi and 40 on Intermittent, compared with three on RAF and

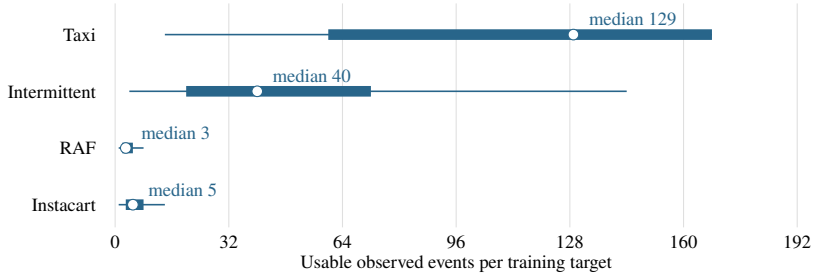


Fig. 3. Available training history under the frozen input windows. Thin lines span the 5th–95th percentiles, thick lines the 25th–75th percentiles, and circles mark medians. These target-weighted distributions characterize input availability, not model performance.

five on Instacart. Thus an equal nominal sequence cap would not imply equal evidence per prediction. These training distributions describe the input populations; they neither measure which events a trained model retains nor determine its forecasting error.

E Event-lag association in the training data

For series e and event lag ℓ, pair $x_{e,j}^{(\ell)} = \log(1 + q_{e,j-\ell})$ with $y_{e,j}^{(\ell)} = \log(1 + q_{e,j})$ whenever both events belong to that training series. Each lag counts observed events in sequence order. It does not denote elapsed calendar time. The descriptive calculation includes the complete training sequence, without the model's recorded-time window or sequence-length cap.

Let $\bar{x}_e^{(\ell)}$ and $\bar{y}_e^{(\ell)}$ be the separate means of the predecessor and successor members of those pairs. The statistic is

$$r_\ell = \mathrm{Corr}\left(x_{e,j}^{(\ell)} - \bar{x}_e^{(\ell)},\ y_{e,j}^{(\ell)} - \bar{y}_e^{(\ell)}\right), \tag{13}$$

where correlation pools the centered pairs across series. Centering removes between-series differences in mean log quantity, but each pair still contributes to the pooled sums. This statistic is consequently different from an average of series-specific correlations. The lag-one column corresponds to the adjacent log correlation in Table 11.

Taxi's association declines with event lag, while Intermittent retains positive association over the displayed range. RAF supplies comparatively few long-lag pairs, and Instacart's correlations remain close to zero. Counts in Figure 4 matter because the available series and pairs change with lag. A series with one pair contributes zero to both centered sums of squares. Constant pairs also provide no variation, and a zero pooled denominator leaves the correlation undefined. RAF's negative values therefore describe these centered samples without establishing a general law of negative dependence.

An archived sensitivity calculation restricts every lag to the same series with at least 20 training events. This cohort retains 131 Taxi series, 4,968 Intermittent series, and 31,020 Instacart series; RAF supplies none. Instacart's lag-one and lag-two correlations

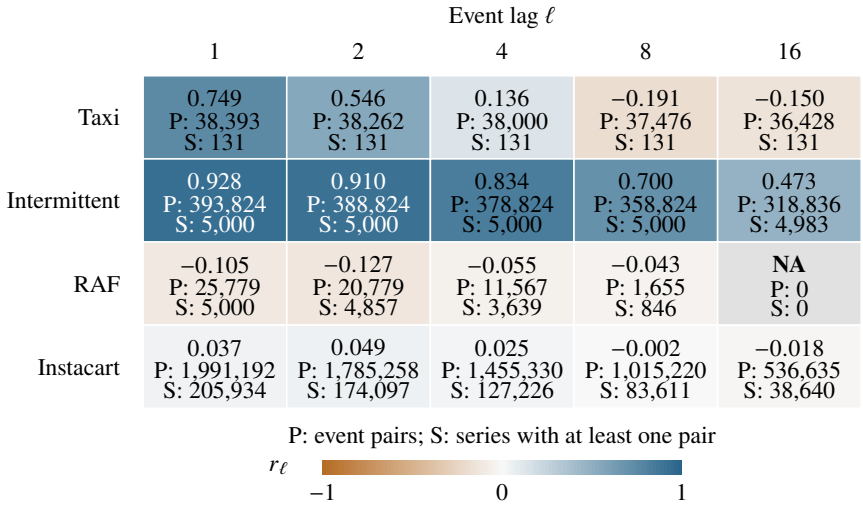


Fig. 4. Train-only within-series log-quantity correlation at event lags 1, 2, 4, 8, and 16. Each cell reports r_ℓ, the number of event pairs (P), and the number of contributing series (S). All cells share the color scale $[-1, 1]$; gray NA denotes no available pairs. Pair counts include constant and single-pair series, which contribute zero to one or both centered sums of squares. Pairs cover complete training sequences without input-window or sequence-cap restrictions.

increase to 0.074 and 0.092 in this cohort. The change cautions against treating the full-population profile as invariant to series composition.

Event-lag association provides a descriptive motivation for examining local and recurrent history paths. It does not measure the contribution of either path. CNN+GRU54 processes contextual states within the retained input window, whereas Eq. (13) describes unconditional log-quantity pairs across complete training sequences. Neither the sign nor the magnitude of r_ℓ establishes a correction gain, a preferred recurrent width, or an architectural ranking. Those claims depend on the matched validation comparisons specified in the main text.

F Diagnostic partitions and aggregation

F.1 Quantity and usable-history partitions

Stratification distinguishes a change in errors within a group from a change in that group's contribution to the aggregate. The quantity boundaries are the training-derived 50th, 90th, 95th, and 99th percentiles, namely $(7, 686, 1562, 3449)$ for Taxi, $(2, 31, 46, 187)$ for Intermittent, $(2, 30, 60, 200)$ for RAF, and $(8, 20, 25, 35)$ for Instacart. Each upper boundary belongs to the lower bin, so the last bin agrees with the strict large-quantity threshold in Table 1.

The stored history partitions divide Taxi, Intermittent, and RAF at 64 and 128 usable events, and Instacart at 1, 3, 7, 15, and 31. They permit within-dataset diagnosis, although

they do not define identical groups across datasets. Their resolution also depends on the sequence cap. Empty strata remain explicit and receive no estimated error. For a fixed dataset, every comparison arm and seed must share target membership and partition boundaries.

All strata inherit the checkpoint selected by minimum overall validation quantity RMSE. Neither a favorable bin nor a timing score changes that selection. A prespecified control remains fixed across bins; if an external comparator is selected by aggregate validation RMSE, that same comparator also remains fixed. These rules prevent a separate best-model choice within each subgroup. The definitions below specify the diagnostic procedure and introduce no additional CNN+GRU performance results.

F.2 Per-seed errors and additive contributions

For model m, seed s, and a nonempty stratum b containing N_b targets, define

$$\mathrm{SSE}_{m,s,b} = \sum_{i \in b} (\widehat{q}_{m,s,i} - q_i)^2, \tag{14}$$

$$\mathrm{MAE}_{m,s,b} = \frac{1}{N_b} \sum_{i \in b} |\widehat{q}_{m,s,i} - q_i|, \qquad \mathrm{RMSE}_{m,s,b} = \sqrt{\frac{\mathrm{SSE}_{m,s,b}}{N_b}}. \tag{15}$$

Compute each metric within a seed before averaging over $\mathcal{S} = \{42, 52, 62\}$. Sample standard deviations describe the dispersion of these seed-level metrics. The count N_b refers to targets in one seed, rather than repeated copies of the same target across seeds.

Let M denote the focal model, C its fixed comparator, and $N = \sum_b N_b$. The contributions of bin b to the aggregate differences are

$$C_b^{\mathrm{MSE}} = \frac{1}{|\mathcal{S}|N} \sum_{s \in \mathcal{S}} (\mathrm{SSE}_{M,s,b} - \mathrm{SSE}_{C,s,b}), \tag{16}$$

$$C_b^{\mathrm{MAE}} = \frac{N_b}{N|\mathcal{S}|} \sum_{s \in \mathcal{S}} (\mathrm{MAE}_{M,s,b} - \mathrm{MAE}_{C,s,b}). \tag{17}$$

Summing these contributions across bins recovers the mean of the seeds' overall MSE or MAE differences. Negative values favor M. The MSE sum is not the difference between squared mean RMSEs, and binwise RMSE differences do not supply an additive decomposition. Empty bins contribute zero to the aggregate through their zero target count, without an estimated bin metric.

Because each contribution includes target share, a small group with large errors can offset gains across a larger group. Reporting both within-bin errors and contributions distinguishes these cases without attributing the difference to a single data characteristic. A model-specific decomposition requires that model's selected-checkpoint validation records. Historical MLP decompositions cannot substitute for CNN+GRU54 results.

G Completed CNN+GRU54 correction records

This appendix records the separate 27-condition correction campaign. Full, correction bypass, and current-only each contribute seeds 42, 52, and 62 on every core dataset. Selection follows the earliest strictly improving finite minimum of full-validation raw-quantity RMSE. The large subsets contain 79 Taxi, 1,141 Intermittent, and 50 RAF targets, with identical target membership across arms and seeds. Neither time NLL nor large-quantity error determines an additional checkpoint.

Table 12. Selected and final saved epochs for the correction study. F denotes full, B denotes correction bypass, and C denotes current-only. Each entry is selected/final. A100 denotes the preserved completed campaign records, not a common RTX cost baseline. RAF seed 42 full/current-only resumed across physical A100 GPUs.

Dataset	Seed	F	B	C	Execution
Taxi	42	126/166	141/181	82/122	A100
	52	74/114	68/108	19/59	A100
	62	84/124	92/132	43/83	A100
Intermittent	42	15/55	30/70	6/46	RTX 5090
	52	11/51	13/53	56/96	RTX 5090
	62	13/53	5/45	38/78	RTX 5090
RAF	42	9/49	9/49	9/49	A100 → A100 (F/C)
	52	5/45	5/45	43/83	A100
	62	20/60	26/66	22/62	RTX 5080

Full and current-only each contain 25,308 active correction parameters. Bypass retains the registered modules but freezes them and does not execute either correction path. The recurrent output projection and local kernels are zero-initialized under the frozen source contract. Fifteen conditions completed on A100 and twelve on RTX. Operational interruptions and checkpoint transport do not create additional scientific conditions. RAF seed 42 full/current-only retained the saved state and selection history through migration. Hardware-segment timings remain separate and concurrent fit wall time is not a structural efficiency estimate.

Table 13. Completed correction study on Taxi. Mean ± sample SD over three seeds at the same selected checkpoint for all and large-quantity populations. These campaign values do not replace Table 2.

Arm	Population	Quantity MAE	Quantity RMSE	Time NLL
Full	All	25.6347 ± 1.0220	80.4089 ± 2.9830	1.2721 ± 0.1648
	Large	306.0319 ± 42.7071	394.7608 ± 42.1609	8.061×10^{-16} $\pm 1.396\times10^{-15}$
Correction bypass	All	28.2757 ± 0.4683	87.6117 ± 1.4166	0.9544 ± 0.1116
	Large	315.6244 ± 3.7287	402.0961 ± 10.2204	4.214×10^{-16} $\pm 7.298\times10^{-16}$
Current-only	All	28.1968 ± 0.3279	88.0008 ± 1.3758	0.8206 ± 0.2261
	Large	303.2394 ± 24.8923	386.7095 ± 30.9643	1.669×10^{-24} $\pm 2.891\times10^{-24}$

Table 14. Completed correction study on Intermittent. Mean ± sample SD over three seeds at the same selected checkpoint for all and large-quantity populations. These campaign values do not replace Table 2.

Arm	Population	Quantity MAE	Quantity RMSE	Time NLL
Full	All	0.7057 ± 0.0148	1.6957 ± 0.0423	0.4097 ± 0.1818
	Large	6.4667 ± 0.4364	8.2968 ± 0.6547	9.239×10^{-5} $\pm 1.527\times10^{-4}$
Correction bypass	All	0.7449 ± 0.0276	1.7409 ± 0.0279	0.3430 ± 0.0405
	Large	5.8346 ± 0.5149	7.8753 ± 0.3017	6.101×10^{-7} $\pm 1.031\times10^{-6}$
Current-only	All	0.7532 ± 0.0227	1.7562 ± 0.0574	0.3479 ± 0.0244
	Large	6.4383 ± 0.1763	8.2016 ± 0.2010	5.550×10^{-9} $\pm 9.612\times10^{-9}$

Table 15. Completed correction study on RAF. Mean ± sample SD over three seeds at the same selected checkpoint for all and large-quantity populations. These campaign values do not replace Table 2.

Arm	Population	Quantity MAE	Quantity RMSE	Time NLL
Full	All	9.2021 ± 0.0801	34.1711 ± 0.2121	3.5481 ± 0.0634
	Large	280.7816 ± 7.5080	327.6237 ± 6.6540	3.5037 ± 0.2080
Correction bypass	All	9.2676 ± 0.0325	34.2964 ± 0.0846	3.5891 ± 0.1286
	Large	277.9862 ± 7.0989	324.7870 ± 6.8087	3.4519 ± 0.1400
Current-only	All	9.3667 ± 0.1663	34.3109 ± 0.1010	3.4693 ± 0.1006
	Large	276.1454 ± 6.7053	323.6122 ± 6.5116	3.4010 ± 0.0764

G.1 Adverse paired differences

Table 16 records how often full correction has a higher metric than each control. Counts refer to seedwise paired differences, not differences between the rounded means. A timing or large-quantity loss remains part of the result even when overall RMSE improves.

Table 16. Number of adverse differences (full minus control > 0) in the completed correction study. Each control contributes three seed comparisons per dataset.

		All			Large		
Dataset	Control	MAE	RMSE	NLL	MAE	RMSE	NLL
Taxi	Bypass	0	0	3	1	1	2
Taxi	Current-only	0	0	3	2	2	1
Intermittent	Bypass	0	0	1	2	2	2
Intermittent	Current-only	0	0	1	2	2	2
RAF	Bypass	1	1	1	1	1	1
RAF	Current-only	0	1	2	2	2	2

These counts retain the RAF seed-52 RMSE failure against both controls, all six Taxi overall time-NLL losses, and large-quantity losses despite some overall quantity gains. The full six-metric records and first-minimum checkpoint histories are retained in the completed Validation ledger. The criterion in Section 4 is a development consistency rule, not a significance or independent-confirmation test.

H Completed Instacart architecture records

The nine fits compare MLP8, MLP12, and CNN+GRU54 over seeds 42, 52, and 62. The evaluation population contains 503,733 targets and 6,036 large-quantity targets with $q > 35$. Each seed repeats the same target population. Recorded intervals are measured in days, top-coded at 30, and scored with time scale seven. These are recorded integer-bin NLLs, not the older clamped time-loss field. Selection and all six metrics inherit the same full-validation quantity-RMSE checkpoint.

Table 17. Selected and final saved epochs for the completed Instacart architecture study. All three architectures share one GPU within each seed. Cross-seed Runtime differences and unequal parameter budgets remain.

Architecture	Seed	Selected	Final	Execution
MLP8	42	30	70	RTX 5080
	52	15	55	RTX 5090
	62	35	75	RTX 5080
MLP12	42	28	68	RTX 5080
	52	15	55	RTX 5090
	62	35	75	RTX 5080
CNN+GRU54	42	30	70	RTX 5080
	52	15	55	RTX 5090
	62	1	41	RTX 5080

Table 18. Completed Instacart six-metric comparison. Entries are mean ± sample SD over seeds 42, 52, and 62. All and large-quantity rows share the same selected checkpoint within each fit.

Arm	Population	Quantity MAE	Quantity RMSE	Time NLL
MLP8	All	3.9910 ± 0.0028	5.8854 ± 0.0135	2.8075 ± 0.0040
	Large	22.0238 ± 0.4187	24.6355 ± 0.3367	2.8579 ± 0.0451
MLP12	All	3.9906 ± 0.0025	5.8801 ± 0.0169	2.8072 ± 0.0050
	Large	22.1032 ± 0.2647	24.6683 ± 0.2247	2.8559 ± 0.0461
CNN+GRU54	All	3.9872 ± 0.0041	5.8774 ± 0.0095	2.8108 ± 0.0090
	Large	21.7822 ± 0.2505	24.4380 ± 0.2012	2.8476 ± 0.0444

CNN+GRU54 has lower overall and large-quantity RMSE and MAE than MLP8 in each seed. Against MLP12, seed 52 instead favors MLP12 on all four quantity metrics. CNN+GRU54 lowers large-quantity time NLL in every paired comparison, but seed 62 raises overall time NLL against both MLP variants. These exceptions remain in the aggregate and do not support a uniform architecture ranking.

The frozen Instacart science revision differs from the correction-campaign revision, but its 126 source files match the corresponding files in the correction study's 129-file closure, including the CNN+GRU and common-head implementations. Native checks record 89,859, 96,003, and 102,879 trainable parameters for MLP8, MLP12, and CNN+GRU54. Binary checkpoint bytes and complete endpoint records were collected and SHA-verified. No CPU binary re-inference or independent evaluation is inferred from that integrity check.